

ECHELON 2026 · CHALLENGE BRIEF

BOLDR · WATCH E-COMMERCE

Build a Self-Improving Customer Intelligence Engine

Don't build a chatbot.

Turn every customer question into product & marketing intelligence.

Shopify

E-commerce Stack

Email

Support Channel

3 staff

CS Team Size

Open-source

Tool Preference

About Boldr

Boldr is a Singapore-based watch micro-brand selling titanium watches, straps, and watch servicing through Shopify. Their products attract a niche, considered buyer — people who care about materials (titanium grade, alloy composition, BPA-free straps), customisation (engraving, strap compatibility), and after-sales service (battery, regulation, full service). A 3-person customer service team handles all inbound enquiries over email, primarily via Gmail.

Most questions are answerable from existing internal documents — rate cards, product spec sheets, and a growing FAQ updated manually whenever a novel question comes in. But the system has **no memory, no feedback loop, and no intelligence layer**. Every question is answered and forgotten. When a customer asks about BPA-free straps, the team answers and moves on — missing that this is a marketing angle nobody in the watch strap space is exploiting.

ShopifyE-commerce
Platform**Email**Support
Channel**3 staff**Customer
Service Team**Your Tools**Open-source
Preferred

The Problem

Boldr's customer service team answers dozens of enquiries every week. The majority are answerable from existing Knowledge Base documents — but the process is entirely manual. A staff member reads each email, searches internal docs, writes a reply, and moves on. When a question falls outside the Knowledge Base, they escalate, get an answer, and sometimes remember to update the FAQ. Sometimes they don't.

The result is a system that stays static while customer questions evolve. Novel questions — the ones that reveal what buyers *actually* care about — disappear into inboxes rather than feeding back into the product and marketing strategy.

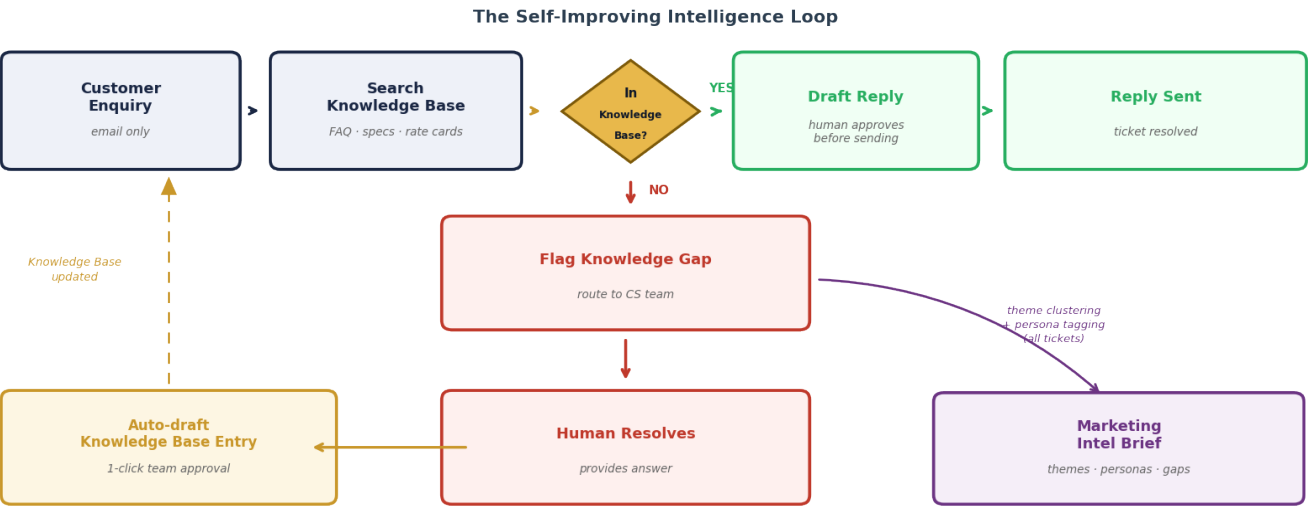
Pain Points	Missed Opportunities
• Every enquiry handled manually — no drafting assist • Knowledge Base updated inconsistently — same questions recur • No visibility on which themes are trending • 3-person team stretched across reply + admin • Zero signal flows from support to marketing	• BPA-free straps: marketing angle nobody is using • Sustainability buyers: growing segment, unaddressed • Nickel allergy queries: product page gap confirmed externally • Health-conscious parents: distinct persona, no targeted messaging • Vegan strap questions: 3 tickets in 6 weeks — new trend forming • Corporate gifting: recurring, no bulk pricing Knowledge Base entry

The Core Insight

When a customer asked "Is this BPA-free?", the team had never been asked before. They checked with the manufacturer, answered, and updated the FAQ. A few weeks later they realised: *this could be a marketing angle nobody in the watch strap space is using*. One support ticket became a product page update and a potential campaign — but only because a human noticed the pattern. **The challenge is to make that happen automatically, at scale.**

The Challenge

Build an AI workflow that transforms Boldr's reactive customer support into a self-improving intelligence engine — one that answers questions, identifies gaps, updates its own knowledge base, and generates marketing signals automatically.



The self-improving loop: enquiries in, drafted replies and marketing intelligence out — Knowledge Base stays current automatically.

Core Challenge: The Intelligence Loop

Step	Action	Detail
1	Ingest enquiry	Receive customer email. Extract question intent and context.
2	Search Knowledge Base	Query Knowledge Base docs, rate cards, FAQ, product specs for a relevant answer.
3	Answerable? Draft reply	If YES: draft a reply in Boldr's brand voice. Queue for human approval before sending.
4	Not answerable? Flag gap	If NO: flag as knowledge gap. Route to CS staff with context. Do NOT hallucinate.
5	Auto-draft Knowledge Base entry	Once human resolves the gap, auto-draft a new Knowledge Base entry in Boldr's format for 1-click approval.
6	Theme clustering	Weekly: group novel questions by theme (materials, sustainability, sizing, gifting).
7	Marketing brief	Monthly output: "What customers are asking that is not on your product pages" — with buyer persona tags.

Buyer Personas to Identify

The workflow must tag each enquiry against one of five buyer personas derived from the sample ticket data. Theme clustering feeds directly into persona identification — which in turn drives the marketing brief output.

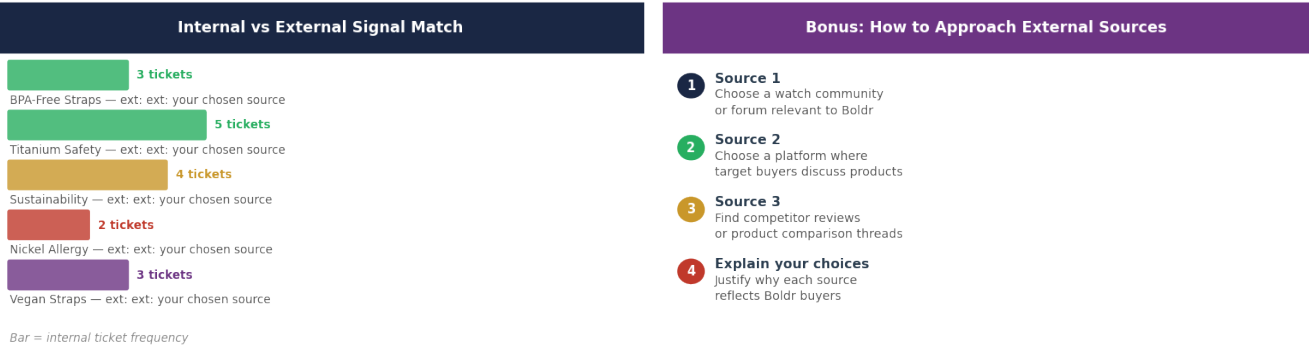
Buyer Personas to Identify & Tag (from sample data)



Five buyer personas identified from real ticket data — each with distinct trigger keywords and marketing actions.

Bonus Challenge: External Sentiment Benchmarking

The best solutions do not stop at internal data. Pair Boldr's internal customer signals against external market sentiment from watch forums, Reddit communities, and competitor reviews — to answer: "Is this a Boldr-specific concern, or a market-wide signal we can capitalise on?"



Internal ticket frequency vs. external forum sentiment — cross-validate on at least 3 themes.

Bonus Requirement	Detail
Identify 2+ external sources	Identify at least two external sources relevant to Boldr's customer base. Explain why each source is appropriate and what buyer signals it captures.
Compare on 3+ themes	Demonstrate how internal customer signals (from ticket data) align or diverge from external market sentiment on at least 3 themes from the ticket data.
Actionable insight output	For each theme: "Is this a Boldr-specific gap or a market-wide concern? What should Boldr do about it?"

Sample Data Provided

Eleven files are provided covering the full data landscape builders need. The customer tickets are the primary input; all other files form the knowledge base the agent must search, update, and generate outputs from.

File	What It Contains	Role in Workflow
01_customer_tickets.csv	Anonymised inbound tickets — subject, body, theme, buyer persona tag, answerable flag	Primary INPUT — what the agent processes
02_product_specs.json	Full product catalogue: SKUs, materials, lug widths, water resistance, strap compatibility	Knowledge Base source — product questions
03a_rate_card_engraving.csv	Engraving pricing, character limits, font options, turnaround times	Knowledge Base source — engraving questions
03b_rate_card_servicing.csv	Servicing tiers, pricing, turnaround times for all models	Knowledge Base source — servicing questions
04_faq_document.pdf	28 existing FAQ entries across all themes — the current Knowledge Base	Knowledge Base source — primary knowledge store
05_decision_tree.json	16 routing rules: trigger conditions, YES/NO branches, escalation flags	Logic layer — classification & routing
06_shipping_reconciliation.csv	Shipping data (bonus context — see Shipping challenge brief)	Reference only
07_knowledge_gap_log.csv	Gaps already identified: theme, frequency, marketing signal flag, Knowledge Base draft status	Target OUTPUT format for gap detection
08_buyer_personas.csv	5 personas: trigger keywords, recommended messaging, marketing opportunity per persona	Target OUTPUT format for persona tagging
09_external_sentiment_data.csv	4 external forum/review sources with sentiment and themes	Bonus INPUT — external benchmarking

Platform Guidance

Participants are free to use any automation platform or framework. Open-source and low-code tools (such as n8n, Make.com, or similar) are encouraged as they promote transparency and maintainability — but the choice is yours. What matters is that the workflow is well-architected, explainable, and something the Boldr team could actually use and own.

Why This Scales Beyond Boldr

Every product brand with a considered buyer and a growing FAQ faces this exact problem. The workflow built here — classify, answer, detect gap, update Knowledge Base, surface themes — is directly replicable across specialty food, cosmetics, outdoor gear, or any Shopify brand where product knowledge is deep and support volume is moderate.